5. Real-Time Fraud Detection System:

- Implement the trained model as a service:
    - Use Flask or FastAPI to create a REST API for model predictions.
    - Process transactions in real-time to flag suspicious activities.
- Integrate the system with alert mechanisms:
    - Send email or SMS notifications to users and administrators.
    - Implement a dashboard to review flagged transactions.

6. Scalability and Maintenance:

- Deploy the system on scalable platforms like AWS or Google Cloud.
- Use load balancers and distributed processing to handle high transaction volumes.
- Continuously retrain the model with new data to adapt to evolving fraud patterns.
- Monitor system performance and add features based on user feedback.

Outcome:

The fraud detection system will effectively identify potentially fraudulent transactions, enabling financial institutions to mitigate risks and enhance security. Real-time capabilities ensure swift responses, reducing financial losses and improving user confidence.

Example Code:

```
import pandas as pd
import numpy as np
from sklearn.ensemble import IsolationForest from sklearn.model_selection import train_test_split
from sklearn.metrics import classification_report, roc_auc_score # Step 1: Load Data
data = pd.read_csv('transactions.csv') # Step 2: Preprocess Data
```